Ethan Tran

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Education

University of Waterloo 2025 – 2031
Bachelor of Applied Science in Computer Engineering

Technical Skills

Languages: Python, C, C++, TypeScript, Swift, Java, C#, Verilog, HTML, CSS
Frameworks/Libraries: FastAPI, React, OpenCV, YOLO, SQLite, Unity
Tools/Hardware: Git, Linux, Raspberry Pi, KiCad, Xcode, VS Code, Visual Studio

Experience

Organizer / Software Developer Jun. 2024 – Mar. 2026
HammerHacks

- Hosted **140+ hackers from 17 high schools**, collaborating across **50+ projects** during a 12-hour in-person hackathon at McMaster University.
- Raised **$12,000 in funding from 15 sponsors** as a registered **501(c)(3) nonprofit**, including support from McMaster Engineering, Hack Club, and Inference Labs.
- Designed and deployed the HammerHacks landing page, achieving a **90% participant conversion rate** and managing registrations end-to-end.

Software Developer Lead Sep. 2023 – Jun. 2025
Saltfleet Parliament

- **Led the design and development** of the Saltfleet Commons website with a team of 3 developers.
- Collaborated to ensure effective **UI/UX design** and **front-end development.**

App Developer Feb. 2024 – Jun. 2024
Swift Kids

- Designed and optimized mobile application interfaces using **Swift**, completing **10+ projects** to improve user experience and app performance.
- Collaborated with a **team of 4 developers** to conceptualize, design, and present iOS app solutions.

Projects

Patch | *Raspberry Pi 5, Python, FastAPI, OpenCV, React* Hack the North 2026

- Built a voice-driven search rover for the Huawei OMNI Live Challenge, placing in the **top 10%**; it scans a room, asks the user to confirm a match, then drives to the object.
- Wired a **Raspberry Pi 5** to a **TB6612FNG** H-bridge and two gear motors on a separate battery supply with shared ground, driving 1 kHz PWM with per-wheel scaling and a 30% minimum command so the motors never stall.
- Implemented **proportional differential steering** from bounding-box error with deadzone hysteresis, and estimated range from apparent object width using one RGB camera and no depth sensor.
- Kept motion safe around multi-second cloud inference with bounded motor pulses, an independent watchdog, and a mission token that stops late model responses from restarting the wheels; covered by **460+ tests**.

Power Rangers Morpher | *Electronics, CAD, RFID*

- Designed and built a Gundam-themed Power Rangers morpher for GitHub HQ hackathon combining aesthetics with functional electronics.
- **Implemented RFID** card detection that triggers the handheld device's lighting on card insertion.
- Incorporated Gundam-inspired wings in the CAD to enhance the visual and mechanical design.

FPGA Digital Control System | *Verilog, Quartus Prime, Model Sim, FPGA*

- Designed an FPGA-based digital control system integrating **combinational/sequential logic, registers, multiplexers, and finite-state control** for real-time hardware operation.
- Implemented SPI communication with an MCP3004 ADC and converted analog potentiometer readings into 50 Hz PWM commands for servo motor control.
- Developed and verified modular RTL components in ModelSim and Quartus Prime, debugging timing and control-signal behavior before deployment to physical FPGA hardware.

TetriPad | *Electronics, Firmware, Embedded Systems*

- Built a custom hackpad for precision Tetris play, with a 4-switch pyramid layout modelled on the classic D-pad.
- Programmed custom **firmware** to handle responsive input and OLED screen feedback.